Yiqing Wang

yq.wang@duke.edu · LinkedIn · Homepage ·

Education

Duke University, Durham, North Carolina, the United States — Aug. 2023 – Present

Ph.D. Candidate Vision and Image Processing Lab, Department of Biomedical Engineering (BME)

- GPA: 3.84/4.0

Shanghai Jiao Tong University (SJTU), Shanghai, China — Sep. 2019 – Jun. 2023

Bachelor's Degree of Eng. major in Biomedical Engineering, minor in Computer Science and Technology

- GPA: 3.84/4.3 (Top 10%)

Scholar Experiences

VIP Lab @ Duke directed by Sina Farsiu — Aug. 2023 – Present

Research Assistant PRIMA: Pre-training with Risk-integrated Image-Metadata Alignment for Medical Diagnosis via LLM

- Transformed clinical metadata into structured semantic representations by fine-tuning ClinicalBERT on RAG-constructed medical corpora, enabling explicit incorporation of domain priors without requiring large-scale paired image–text datasets.
- Proposed a unified multimodal pre-training framework with four complementary loss functions and an end-to-end pipeline leveraging Qwen3 to coordinate global-local feature alignment and multimodal feature synthesis across heterogeneous clinical data.
- Achieved state-of-the-art performance with consistent gains and strong cross-dataset generalization on PAD-UFES-20 and AQUA; abstract accepted as an *Oral Presentation* at *ARVO 2026*, full paper submitted to *MICCAI 2026* and available on arXiv, with an extended journal manuscript in preparation.

Research Assistant A Multi-Granularity Language Learning Approach to Boost Visual Understanding

- Proposed a contrastive learning framework with multi-granular objectives that enables joint multi-label and cross-granularity alignment for medical image understanding, improving representation consistency across hierarchical label structures.
- Constructed large-scale retinal and X-ray image-text datasets to facilitate systematic evaluation across diverse label granularities.
- Demonstrated strong performance across multiple benchmarks; accepted by *ICLR 2026* and available on arXiv.

Research Assistant An Automated Quantitative Ulcer Analysis (AQUA) Algorithm for Microbial Keratitis Organism Classification

- Proposed a contrastive learning-based approach to extract robust representations across heterogeneous clinical data patterns.
- Developed a three-stage multimodal framework integrating imaging, metadata, and clinical features for organism-type classification.
- Journal manuscript in preparation (expected 2026).

CCVL @ JHU & VLAA @ UCSC directed by A. Yuille, Y. Zhou, C. Xie — June 2022 – Nov. 2022

Summer Intern Multi-View MAE for 3D Medical Image Representation Learning (SwinMM)

- Proposed SwinMM, a masked multi-view self-supervised learning framework for data-efficient 3D medical image representation learning, addressing the scarcity of large-scale medical pre-training data.
- Designed a masked multi-view encoder with diverse proxy tasks and a cross-view decoder with cross-attention to exploit cross-view consistency and aggregate latent multi-view information from 3D volumes.
- Achieved superior segmentation performance and improved data efficiency compared to Swin UNETR; published in *MICCAI 2023* (Oral Presentation).

Advanced MRI Lab @ SJTU directed by Hongjiang Wei Feb. 2022 – Jan. 2023

Intern Brain region segmentation and age estimation using QSM

- Proposed STAN, a QSM-based two-stage framework for brain region segmentation and age estimation.
- Integrated segmentation-driven representation learning with transformer-based age prediction to fuse global and local features.
- Achieved accurate brain age prediction on a large cohort; published in *ISMRM 2023* and *IEEE JBHI*.

Publications

2026

Y Wang, C He, MC Lu, M Pawar, L Niziol, M Woodward, S Farsiu. PRIMA: Pre-training with Risk-integrated Image-Metadata Alignment for Medical Diagnosis via LLM. *ArXiv:2602.23297.*

Z Li[1], **Y Wang**[1], S Farsiu, P Kinahan. Boosting Medical Visual Understanding From Multi-Granular Language Learning. *The Fourteenth International Conference on Learning Representations (ICLR).*

Y Wang, C He, LM Niziol, M Pawar, M Lu, MA Woodward, & S Farsiu. Large-Language-Model-Enhanced Multi-Modal Subtype Diagnosis of Microbial Keratitis on Slit-Lamp Photographs and Metadata. *Association for Research in Vision and Ophthalmology (ARVO) 2026 Annual Meeting.*

J Ong, MC Lu, C Thanitcul, M Pawar, JN Hart, EL Vogt, C Deng, S Farsiu, **Y Wang**, P Dmitriev, A Gupta. Deep Learning-Based Classification of Slit-Lamp Photograph Quality in Microbial Keratitis. *Ophthalmology Science.* 2026 Jan 21:101086.

2025

Z Li[1], **Y Wang**[1], S Farsiu, P Kinahan. Large scale data harmonization of radiology studies via multigranular vision language alignment. *SPIE Medical Imaging 2026.*

EL Vogt, LM Niziol, Z Yang, **Y Wang**, M Pawar, P Dmitriev, ... & MA Woodward. Association of deep learning imaging algorithm measures of microbial keratitis with vision outcomes. *Cornea, pp.10-1097.*

J Ong, M Lu, C Thanitcul, M Pawar, JN Hart, E Vogt, S Farsiu, **Y Wang**, ... & MA Woodward. Automated Deep Learning Classification of the Quality of Slit-Lamp Photographs of Microbial Keratitis. *Investigative Ophthalmology & Visual Science*, 66(8), 4436-4436.

Z Yang, MA Woodward, LM Niziol, M Pawar, NV Prajna, A Krishnamoorthy, **Y Wang**, M Lu, S Selvaraj, & S Farsiu. Self-knowledge Distillation-empowered Directional Connectivity Transformer for Microbial Keratitis Biomarkers Segmentation on Slit-lamp Photography. *Medical Image Analysis*, 102, 103533.

2023

M Chen[1], **Y Wang**[1], Y Shi[1], J Feng, R Feng, X Guan, ... & H Wei. Brain Age Prediction Based on Quantitative Susceptibility Mapping Using the Segmentation Transformer. *IEEE Journal of Biomedical and Health Informatics*, vol. 28, no. 2, pp. 1012-1021.

Y Wang[1], Z Li[1], J Mei[1], Z Wei[1], L Liu, C Wang, ... & Y Zhou. SwinMM: Masked Multi-view with Swin Transformers for 3d Medical Image Segmentation. *2023 International Conference on Medical Image Computing and Computer-Assisted Intervention* (pp. 486-496).

Y Wang, Y Shi, H Wei. A Brain Age Estimation Network based on QSM using the Segment Transformer. *2023 International Society for Magnetic Resonance in Medicine (ISMRM).*

Honor and Awards

Oral Presentation at ARVO 2026	May. 2026
Imaging Informatics Best Paper Award Runner Up at SPIE Medical Imaging 2026	Feb. 2026
Oral Presentation at MICCAI 2023	Oct. 2023
Outstanding Graduate of Shanghai Jiao Tong University (Top 5%)	Jun. 2023
Scholarship of School of Biomedical Engineering Alumni Association	Nov. 2022
Merit Student of Shanghai Jiao Tong University (Top 3%)	Oct. 2022
Shanghai Municipal Government Scholarship (Top 1%)	Oct. 2021
Class A Scholarship of Shanghai Jiao Tong University (Top 1%)	Oct. 2020

Service

Conference Reviewer MICCAI 2025; MICCAI 2024;

Journal Reviewer Image and Vision Computing; IEEE Journal of Biomedical & Health Informatics (JBHI);